EGGS AND NESTING

Although many birds lay their eggs directly onto the ground or some other surface, most build a structure, known as a nest, in which to put them. Nests provide safety, insulation, and a fixed point for the adults to concentrate on nurturing their eggs and young. An important part of nest-building is choosing a suitable site—one that will provide concealment or inaccessibility from predators. Nests are made from various materials, depending on what is available—usually vegetation, but also animal hair, feathers, or even shed snake skin or human artifacts.

PROTECTIVE NEST
The most common type of nest is an open cup. Most nests are made of more than one material: this chaffinch's nest is bound by spiders' webs and built up with moss, lichen, grass, and feathers.

BLACK-HEADED TROGON

SOUTHERN BOUBOU

NATAL CHAT

CETTI'S WARBLER

BLUE SHORTWING

EGG COLOR
Birds' eggs have a wide range of colors. However, all shell colors are the product of just 2 pigments, one derived from hemoglobin and the other from bile. These are added as the egg moves down the female's urogenital tract.

between species, and some birds produce more than a single clutch in a year.

A bird's egg is contained by a light but strong shell that protects the developing embryo and acts as a barrier against bacteria. The shell is made of calcium carbonate, which the female absorbs from her food. Despite appearing to be hard, the shell is porous, allowing for the free exchange of oxygen and carbon dioxide across its surface. Inside, the embryo is nourished by a large reservoir of nutritive material that promotes its growth. Even so, when young birds hatch out, they are often poorly developed. In most species, the hatchlings are blind and without feathers; they also lack the ability to regulate their internal temperature, so they must be brooded. These dependent chicks are referred to as nidicolous or altricial young and are completely reliant on their parents for warmth and food. In contrast, some groups of birds, such as waterfowl and gamebirds, hatch nidifugous or precocial young, which are covered with down and are able to feed themselves within hours of hatching.

EURASIAN BLUE TIT HATCHLINGS

DUCK HATCHLING

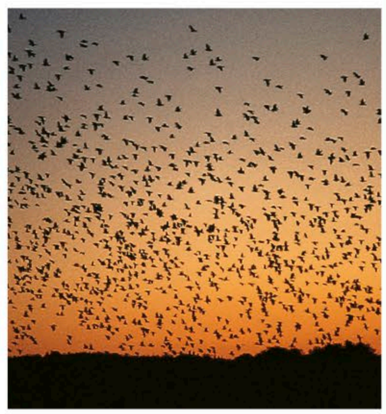
THE FIRST DAYS OF LIFE
Most young birds, including passerines (such as the Eurasian blue tit), depend on adults for food and warmth. Their eggs are laid in well-protected nests. Waders, waterfowl, and gamebirds produce more independent young that can walk and fend for themselves within hours.

Social groups

Birds vary greatly in the way that they relate to their own kind. Some are solitary or form pairs within a defined territory. Many meet up with others for specific activities, such as roosting, feeding, or breeding. Others live in groups throughout their lives.

A social lifestyle carries both advantages and disadvantages. Birds in groups benefit from the collective effort put into searching for food, particularly where supplies are sparsely distributed. Birds in flocks are also less at risk from predators: to be one among many reduces the chances of being singled out and caught, and individuals are more likely to have early warning of a predator's approach. This state of collective vigilance allows birds in a group to spend more time feeding or sleeping than they would if they were alone. In addition, some birds huddle together while roosting to share body heat and reduce heat loss. However, there are also disadvantages in being part of a flock. If food is scarce, competition between group members can cause weaker individuals to become marginalized, unable to feed properly and vulnerable to predators; such birds would be better off on their own. Another disadvantage of social living is the threat of diseases, which spread more rapidly through a group than they do among a population of solitary individuals.

COMMUNAL FEEDING
Flamingos are among the most sociable of all birds, living in vast groups (sometimes flocks of a million or more) throughout their lives and performing all living functions in company with others. Even their courtship displays are communal, synchronized affairs.



FLOCKING TOGETHER
European starlings feed and roost together, and most also breed in small colonies. Starling roosts are famous for their spectacular aerial maneuvers prior to settling down. Up to 2 million birds may share the same roost site.

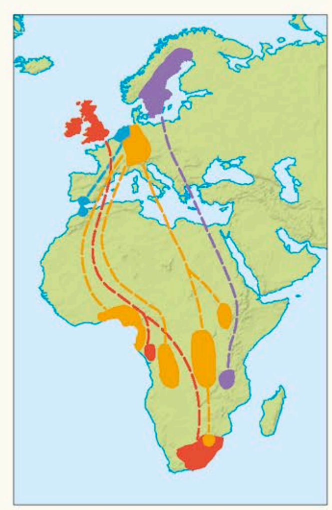
Migration

Many birds undertake seasonal migrations. Most migratory species breed in spring and summer in high latitudes, taking advantage of the relatively long days, but move in winter to lower latitudes, where temperatures are higher. Many species show complex patterns of movement, with only certain populations—or in some cases, just females—leaving the breeding areas while others remain behind. Birds that behave in this way are referred to as partial migrants. Not all birds migrate, partly because migration consumes a lot of energy; those that do not migrate at all (including many tropical species) are referred to as sedentary.

The urge to migrate is triggered by a combination of internal physiological cycles (such as hormone levels) and changes in day length. As the time to migrate approaches, birds lay down reserves of fat to sustain them on their journey and show signs of restlessness. Some birds have an impressive ability to navigate, traveling thousands of miles to arrive at a destination with pinpoint accuracy (see panel, below).

MIGRATION ROUTES

Birds have various ways of navigating. They orient themselves mainly using their efficient internal body clock, which enables them to measure changes in day length, as well as the position of the sun and, at night, of the moon and stars. Many species, if not all, can also detect variations in Earth's magnetic field, which they use as a compass. Birds that have already made several migratory journeys recall landmarks and may use clues such as smells or ultrasounds.



LONG-DISTANCE MIGRANTS
Swallow populations that winter in different parts of Northern Europe follow a range of migration routes. Some travel the relatively short distance to southern Spain or North Africa. Others cover the length of the African continent.